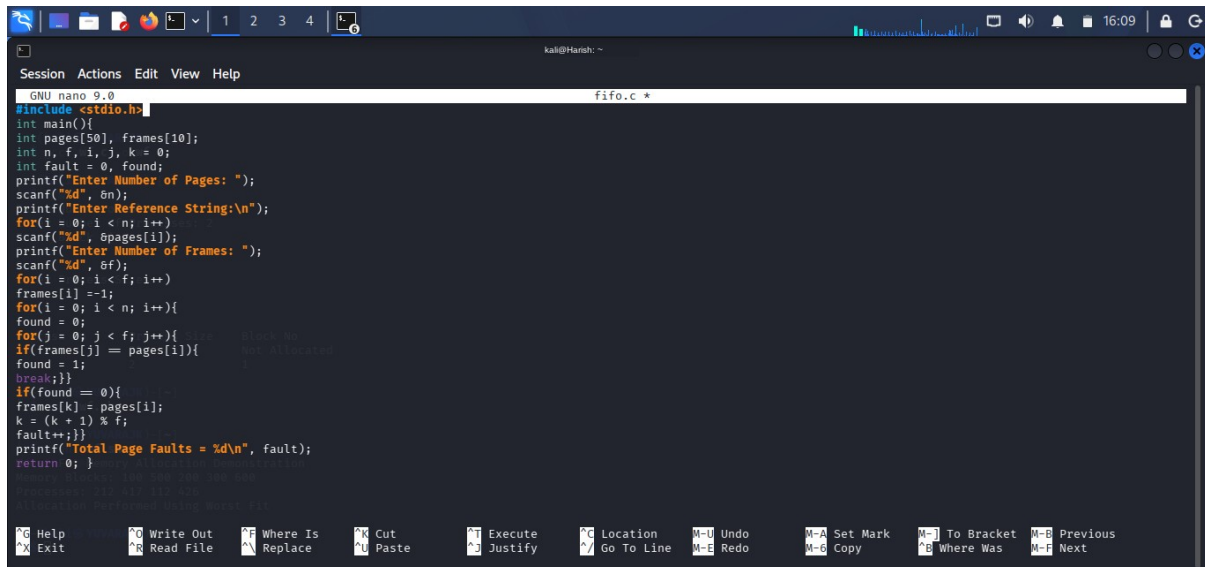


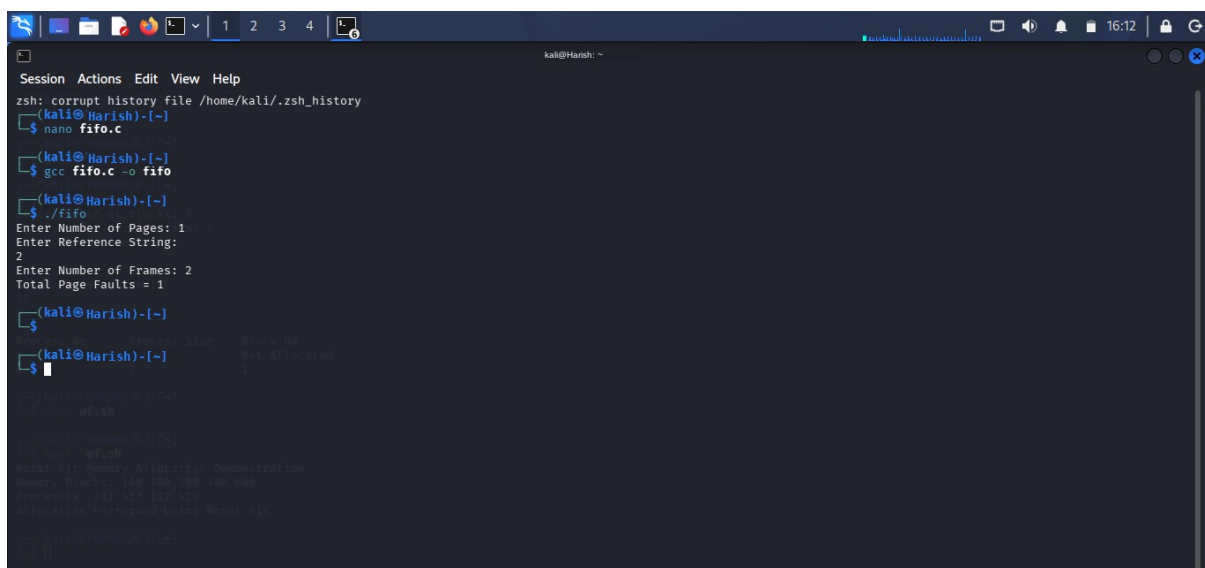
# C PROGRAMMING : FIFO



```
GNU nano 9.0 fifo.c *
#include <stdio.h>

int main(){
    int pages[50], frames[10];
    int n, f, i, j, k = 0;
    int fault = 0, found;
    printf("Enter Number of Pages: ");
    scanf("%d", &n);
    printf("Enter Reference String:\n");
    for(i = 0; i < n; i++){
        scanf("%d", &pages[i]);
    }
    printf("Enter Number of Frames: ");
    scanf("%d", &f);
    for(i = 0; i < f; i++){
        frames[i] = -1;
    }
    for(i = 0; i < n; i++){
        found = 0;
        for(j = 0; j < f; j++){
            if(frames[j] == pages[i]){
                found = 1;
                break;
            }
        }
        if(found == 0){
            frames[k] = pages[i];
            k = (k + 1) % f;
            fault++;
        }
    }
    printf("Total Page Faults = %d\n", fault);
    return 0;
}
```

## OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ nano fifo.c
kali@kali:~$ gcc fifo.c -o fifo
kali@kali:~$ ./fifo
Enter Number of Pages: 1
Enter Reference String:
2
Enter Number of Frames: 2
Total Page Faults = 1
kali@kali:~$
```

# SHELL PROGRAMMING : FIFO

[illegible]

## OUTPUT :

```

kali@Harish: ~
Session Actions Edit View Help
zsh: corrupt_history file /home/kali/.zsh_history
kali@Harish)-[~]
$ nano fifo.c
kali@Harish)-[~]
$ gcc fifo.c -o fifo
kali@Harish)-[~]
$ ./fifo
Enter Number of Pages: 1
Enter Reference String:
2
Enter Number of Frames: 2
Total Page Faults = 1
kali@Harish)-[~]
$
kali@Harish)-[~]
$ nano fifo.sh
kali@Harish)-[~]
$ bash fifo.sh
FIFO Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
Frames: 3
FIFO Algorithm Executed on Demonstration
10 Page Faults
10 Page Faults Using Worst Fit

```

# C PROGRAMMING :LRU

```
GNU nano 9.0 lru.c *
#include <stdio.h>
int main() {
    int pages[50], frames[10], time[10];
    int n, f, i, j;
    int fault = 0, count = 0;
    int found, pos, min;
    printf("Enter Number of Pages: ");scanf("%d", &n);
    printf("Enter Reference String:\n");
    for(i = 0; i < n; i++)scanf("%d", &pages[i]);
    printf("Enter Number of Frames: ");scanf("%d", &f);
    for(i = 0; i < f; i++)frames[i] = -1;
    for(i = 0; i < n; i++){found = 0;
    for(j = 0; j < f; j++){
        if(frames[j] == pages[i]){count++;time[j] = count;
        found = 1;break;}}
    if(found == 0){
        min = time[0];
        pos = 0;
        for(j = 0; j < f; j++){
            if(frames[j] == -1){pos = j;break;}
            if(time[j] < min){
                min = time[j];
                pos = j;}}
        frames[pos] = pages[i];
        count++;
        time[pos] = count;
        fault++;}
    printf("Total Page Faults = %d\n", fault);
    return 0;
}
```

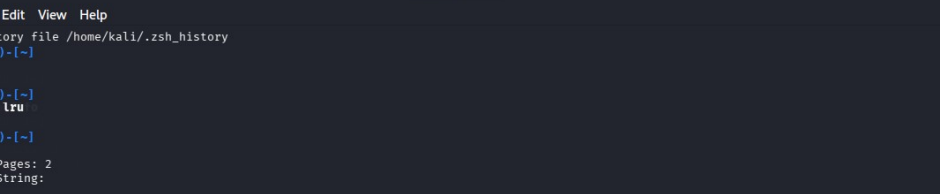
## OUTPUT :

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@Harish)-[~]
$ nano lru.c
(kali@Harish)-[~]
$ gcc lru.c -o lru
(kali@Harish)-[~]
$ ./lru
Enter Number of Pages: 2
Enter Reference String:
5
Enter Number of Frames: 1
Total Page Faults = 2
(kali@YUVARAJK)-[~]
$
$ cat file.sh
#!/bin/bash
# Page Replacement Demonstration
# Reference String: 5 4 1 2 6 3 5 4 2 1 5 4 2
# Frames: 1
# LRU Algorithm Selected
# Author: YUVARAJK
$ ./file.sh
5 4 1 2 6 3 5 4 2 1 5 4 2
Frames: 1
LRU Algorithm Selected
Total Page Faults = 2
```

# SHELL PRROGRAMMING : LRU

[illegible]

## OUTPUT :



The screenshot shows a Kali Linux terminal window with the following content:

```
kali@Harish: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@Harish)-[~]  
$ nano lru.c  
(kali@Harish)-[~]  
$ gcc lru.c -o lru  
(kali@Harish)-[~]  
$ ./lru  
Enter Number of Pages: 2  
Enter Reference String:  
5  
3  
Enter Number of Frames: 1  
Total Page Faults = 2  
(kali@Harish)-[~]  
$ nano lru.sh  
(kali@Harish)-[~]  
$ bash lru.sh  
LRU Page Replacement Demonstration  
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2  
LRU Algorithm Executed  
(kali@Harish)-[~]  
$
```

# C PROGRAMMING :OPTIMAL PAGE

```
GNU nano 9.0 optimal.c *
#include <stdio.h>
int main(){
    int pages[50], frames[10], n, f, i, j, k, pos, fault = 0, found;
    printf("Enter Number of Pages: ");scanf("%d", &n);
    printf("Enter Reference String:\n");
    for(i = 0; i < n; i++){
        scanf("%d", &pages[i]);
    }
    printf("Enter Number of Frames: ");scanf("%d", &f);
    for(i = 0; i < f; i++){
        frames[i] = -1;
    }
    for(i = 0; i < n; i++){
        found = 0;
        for(j = 0; j < f; j++){
            if(frames[j] == pages[i]){
                found = 1;
                break;
            }
        }
        if(found == 0){
            for(j = 0; j < f; j++){
                int future = 999;
                for(k = i + 1; k < n; k++){
                    if(frames[j] == pages[k]){
                        future = k;
                        break;
                    }
                }
                if(j == 0 || future > pos){
                    pos = future;
                }
            }
            frames[0] = pages[i];
            fault++;
        }
    }
    printf("Total Page Faults = %d\n", fault);
    return 0;
}
```

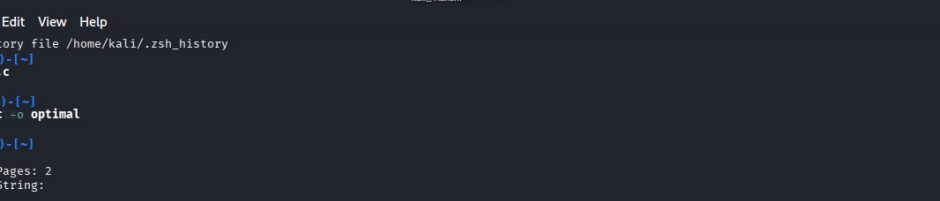
## OUTPUT :

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@Harish)-[~]
$ nano optimal.c
(kali@Harish)-[~]
$ gcc optimal.c -o optimal
(kali@Harish)-[~]
$ ./optimal
Enter Number of Pages: 2
Enter Reference String:
2
3
Enter Number of Frames: 1
Total Page Faults = 2
(kali@YUVARAJK)-[~]
$
```

## SHELL PROGRAMMING :

[illegible]

## OUTPUT :



```
kali@Harish: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@Harish)-[~]  
$ nano optimal.c  
(kali@Harish)-[~]  
$ gcc optimal.c -o optimal  
(kali@Harish)-[~]  
$ ./optimal  
Enter Number of Pages: 2  
Enter Reference String:  
2  
3  
Enter Number of Frames: 1  
Total Page Faults = 2  
(kali@Harish)-[~]  
$ nano optimal.sh  
(kali@Harish)-[~]  
$ bash optimal.sh  
Optimal Page Replacement Demonstration  
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2  
Optimal Algorithm Executed  
(kali@Harish)-[~]  
$
```